Diseases of chrysanthemum

1. Septoria blotch - *Septoria chrysanthemella*

2. Powdery mildew - *Erysiphe cichoracearm*
3. Cercospora leaf spot - *Cercospora chrysanthemi*

4. Grey mold - *Botrytis cinerea*

5. Phyllosticta leaf spot - *Phyllosticta chrysanthemi*

6. Rust - *Puccinia chrysanthemi, P. horiana*

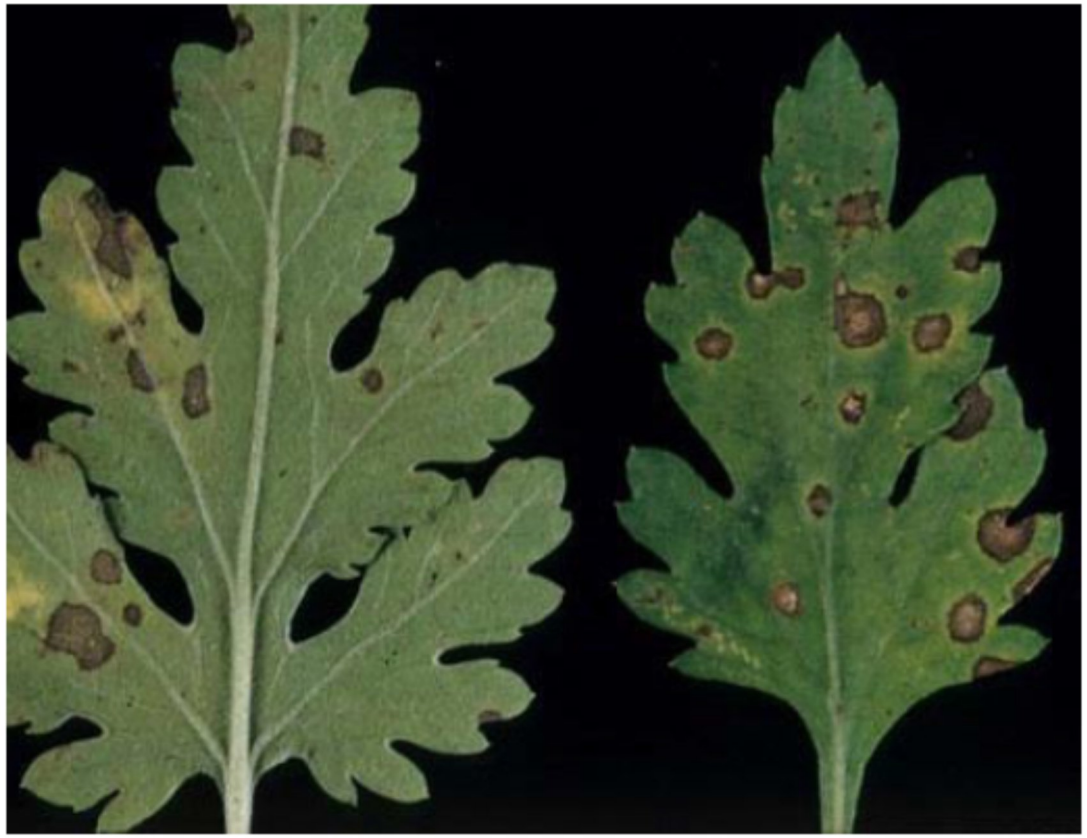
7. Fusarium wilt - *Fusarium oxysporum* f.sp. *chrysanthemi*

8. Bacterial blight - *Erwinia chrysanthemi*

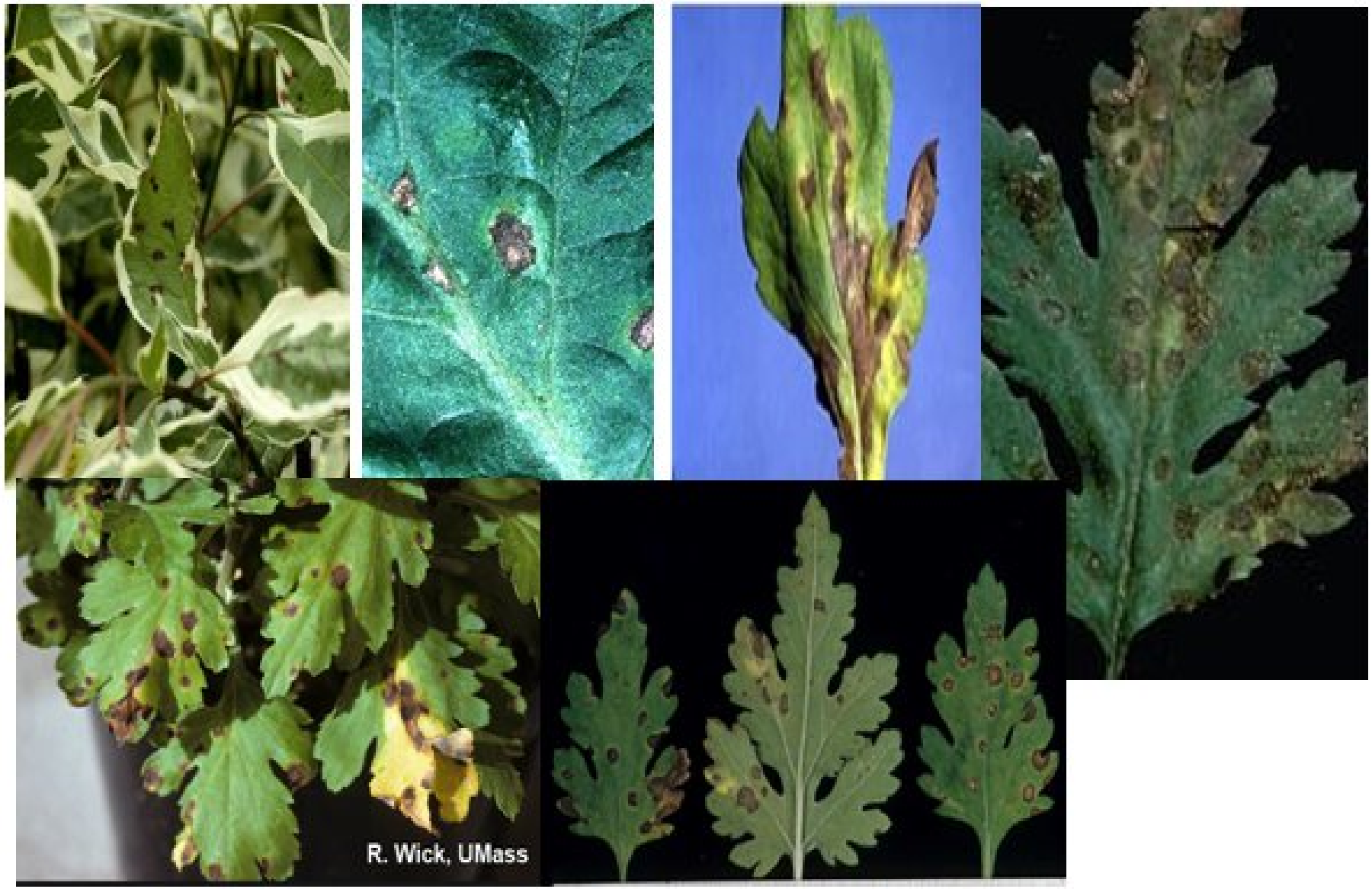
9. Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

10. Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Septoria blotch - *Septoria chrysanthemella*



Septoria blotch - *Septoria chrysanthemella*

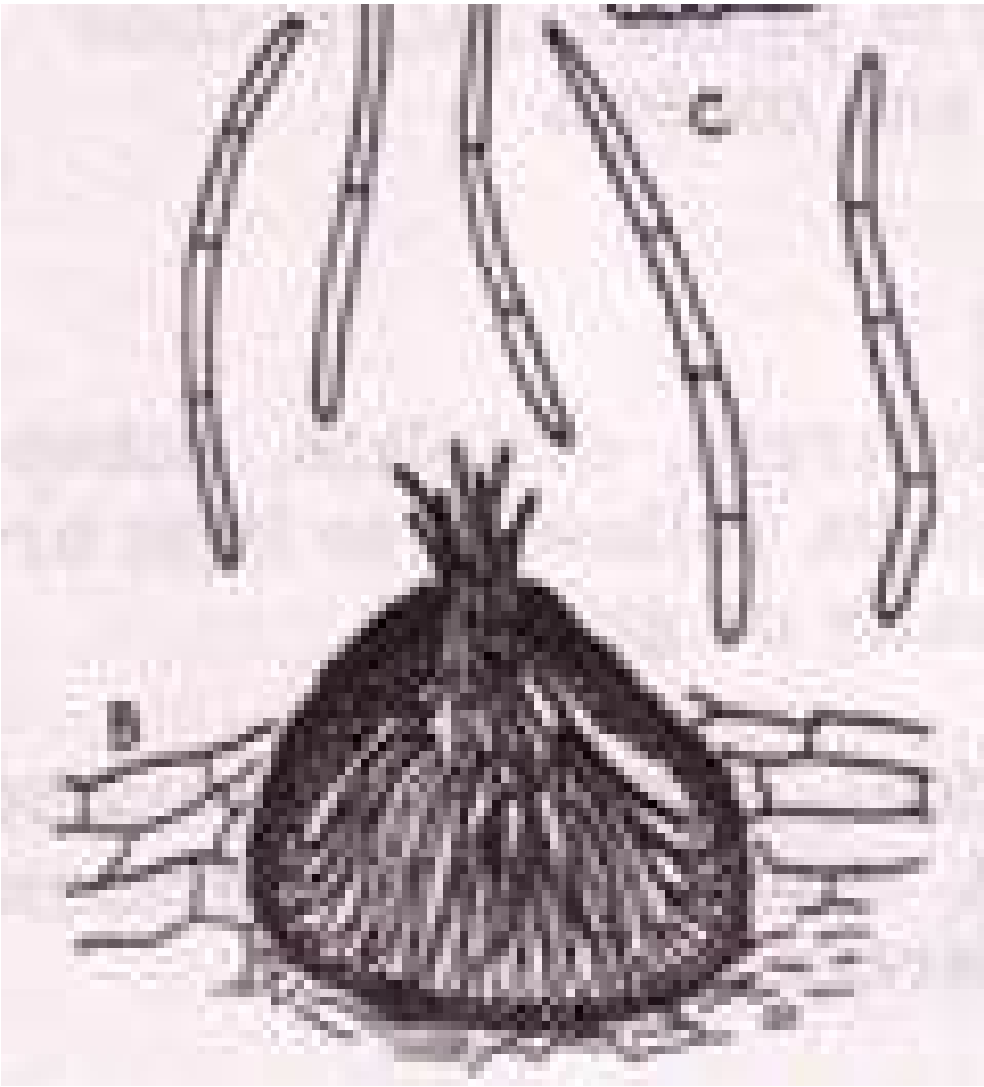


Septoria blotch - *Septoria chrysanthemella*

Symptoms

- **Blackish brown, circular to irregular spots** later form **large patches** covering major portion **of leaf.**
- Patches are surrounded by **chlorotic halo**
- Later the **centers turn greyish** and the leaves remain small and curl
- **Dead leaves hang on stem** for some time

Septoria spp



Survival and spread

- **P.I:** Infected plant debris in soil
- **S.I:** Wind borne conidia

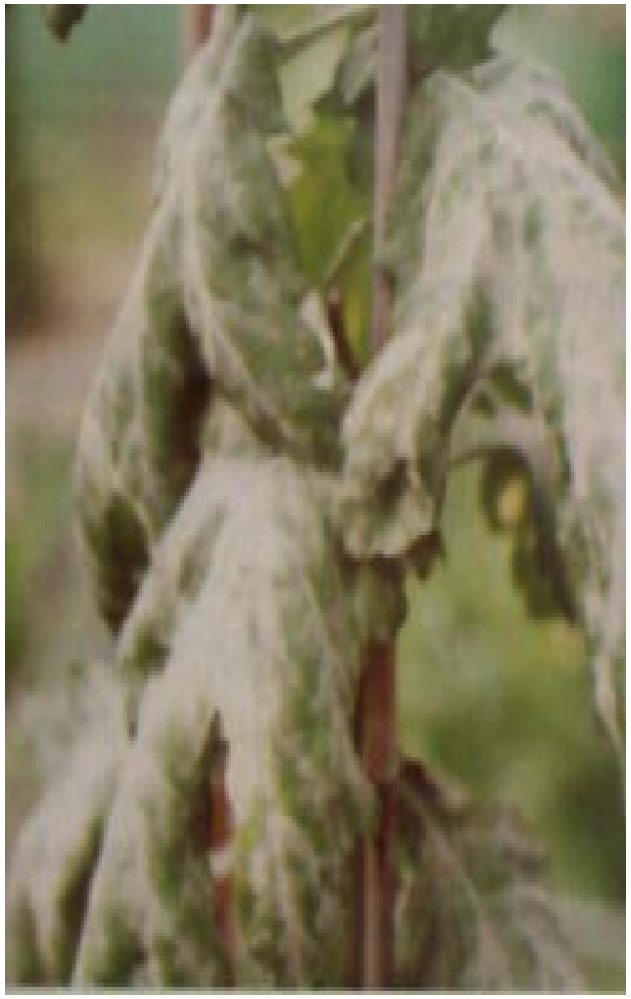
The perfect stage is *Mycosphaerella*

Septoria blotch - *Septoria chrysanthemella*

Management

- Collect and burn diseased plant material
- The application of carbendazim (0.1%) is effective in checking the infection.
- **Excessive irrigation should be avoided**.
- In severe cases, fungicides such as chlorothalonil, mancozeb, myclobutanil, propiconazole, or thiophanate methyl may be applied.
- **Tolerant varieties: Alpana, Sarad and Flirt**

Powdery mildew - *Oidium chrysanthemi* (PS: *Erysiphe cichoracearm*)



Powdery mildew - *Oidium chrysanthemi* (PS: *Erysiphe cichoracearm*)

Symptoms

- As the name implies, it is characterized by a white to **ash- gray powdery growth on leaves and occasionally stems.**
- Foliage may become puckered or distorted; severely infected leaves will shrivel and die.
- The disease is **most serious during hot, humid weather.**
- Unlike most fungal diseases, free water is not required for powdery mildew infection; **high humidity encourages disease development.**

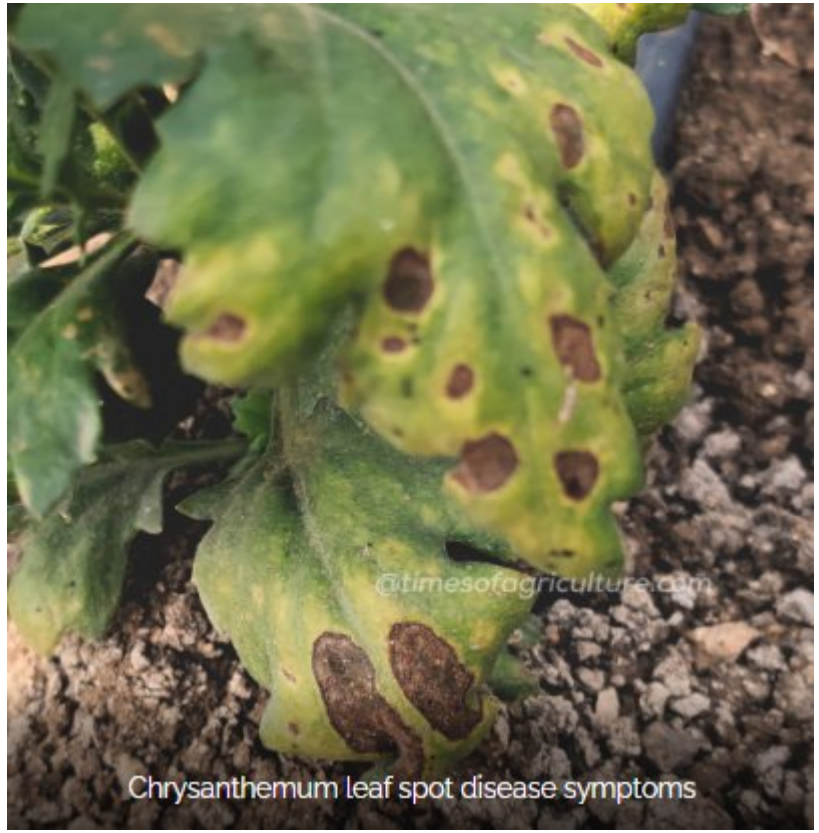
Powdery mildew - *Oidium chrysanthemi* (PS: *Erysiphe cichoracearm*)

Management:

- Powdery mildew can be avoided by proper **plant spacing, good air circulation, low relative humidity** with adequate light.

- Apply preventive fungicides at the first sign of disease with myclobutanil (1.3g/l), triadimefon (0.5 to 1 gm/l), propiconazole (1ml/l), sulphur (2.5g/l), potassium bicarbonate (10g/l), or thiophanate methyl (0.5g/l).

Cercospora leaf spot - *Cercospora chrysanthemi*



Cercospora leaf spot - *Cercospora chrysanthemi*

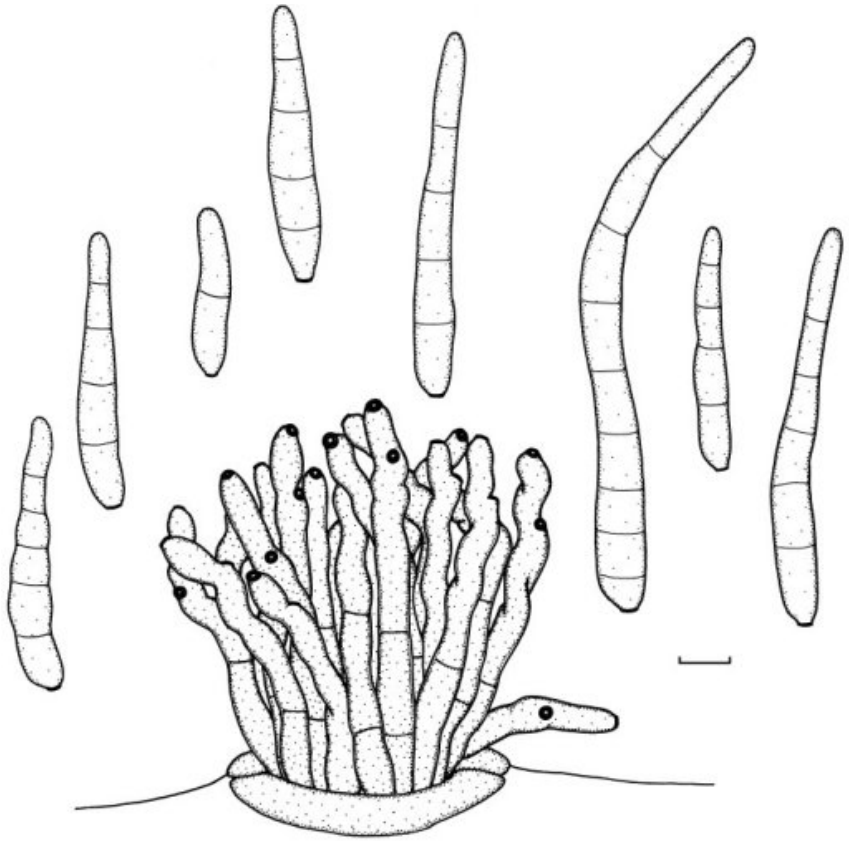
Symptoms

- **Leaf drying is a common symptom in all leaf spot pathogens**.
- **Minute, circular to irregular, brown to dark brown specks** first appear on **leaf tips and margins**, grow bigger and form **spots** with yellow border.

Management

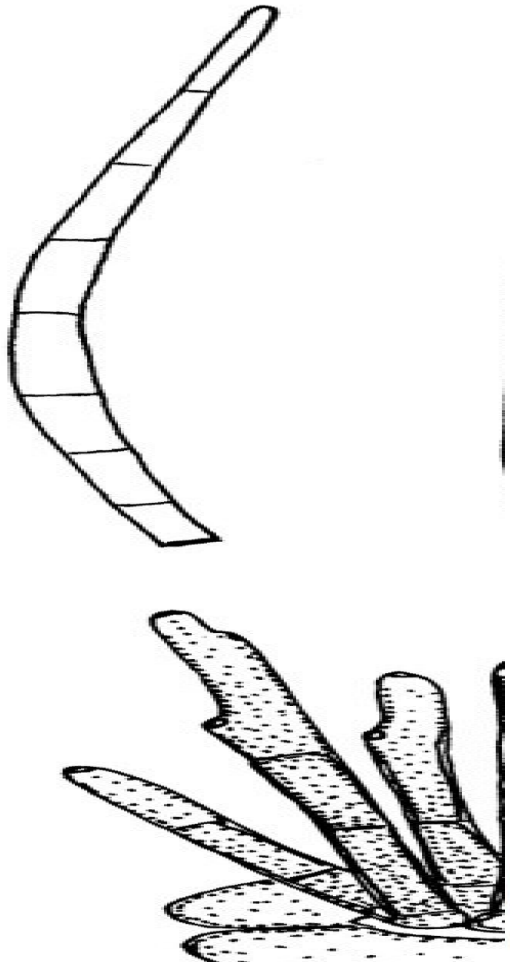
- Destroy the infected leaves
- Avoid Wetting leaves while watering
- Weekly application of mancozeb or zineb 0.2%

Cercospora leaf spot - *Cercospora chrysanthemi*



Cercospora leaf spot - *Cercospora chrysanthemi*

- The fungus produces conidia of vary in size breadth, length ratio of 1:100 - 1:150, slender, straight to slightly curved, multicellular / multiseptate, hyaline or brownish conidia on short dark conidiophores.
- One of the most important features of conidial morphology is the presence a conspicuous, **thickened and dark scar called hilum** which indicates the point of attachment to the conidiophores.



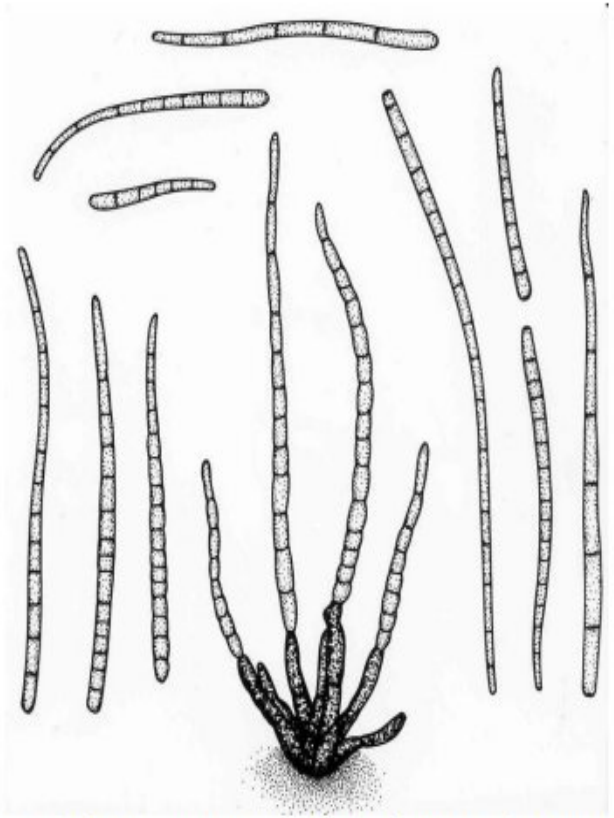


Figure 2. Cercospora carotae, the fungus that causes Cercospora leaf blight or spot of carrot, as it would appear under high power of a light microscope(drawing by Katharine A. Golasyn-Wright)

Cercospora leaf spot - *Cercospora chrysanthemi*

- Teleomorphs of Cercospora as been reported in majority of the cases, Mycospherella known as pseudostromata, or false stromata (conidia-producing hyphae), because they are composed of both fungal tissue and remnants of host tissue (Erikson, 1981).

- Pseudostromata can persist on plant debris for 2 years and have long been regarded as the primary inoculum sources for infection

Phyllosticta leaf spot - *Phyllosticta chrysanthemi*

Phyllosticta leaf spot - *Phyllosticta chrysanthemi*

Symptoms

- It produces numerous **tiny, irregular necrotic spots.**
- The spots are dark coloured later **turning greyish at central portions,** have **distinct margins and 2-3 mm in diameter**.

Phyllosticta leaf spot - *Phyllosticta chrysanthemi*

Management

- Destroying the infected leaves
- Avoid wetting leaves of while watering
- Weekly application of mancozeb or zineb 0.2%

Grey mold - *Botrytis cinerea*

Grey mold - *Botrytis cinerea*

- There are some 50 species of *Botrytis* and infect 200 plant species
- *Botrytis cinerea* has the largest host range than any other *Botrytis* species
- These are necrotrophs
- *Botrytis cinerea* causes **damping-off or seedling blight, leaf blight, stem canker, stem and crown rot, blossom blight, bud rot and cutting rot.**

Grey mold - *Botrytis cinerea*

Symptoms

- Gray mold may occur **on petals, leaves, stem as brown, water-soaked spots.**
- Infected plant parts may be **covered with grayish - brown, powdery masses of spores.**
- **Infected flower buds** do not open
- **Wounded or senescent** tissue are most susceptible
- Gray mold is favored by **extended periods of cloudy, humid, and wet weather.**

Grey mold - *Botrytis cinerea*

PS : *Botryotinia*

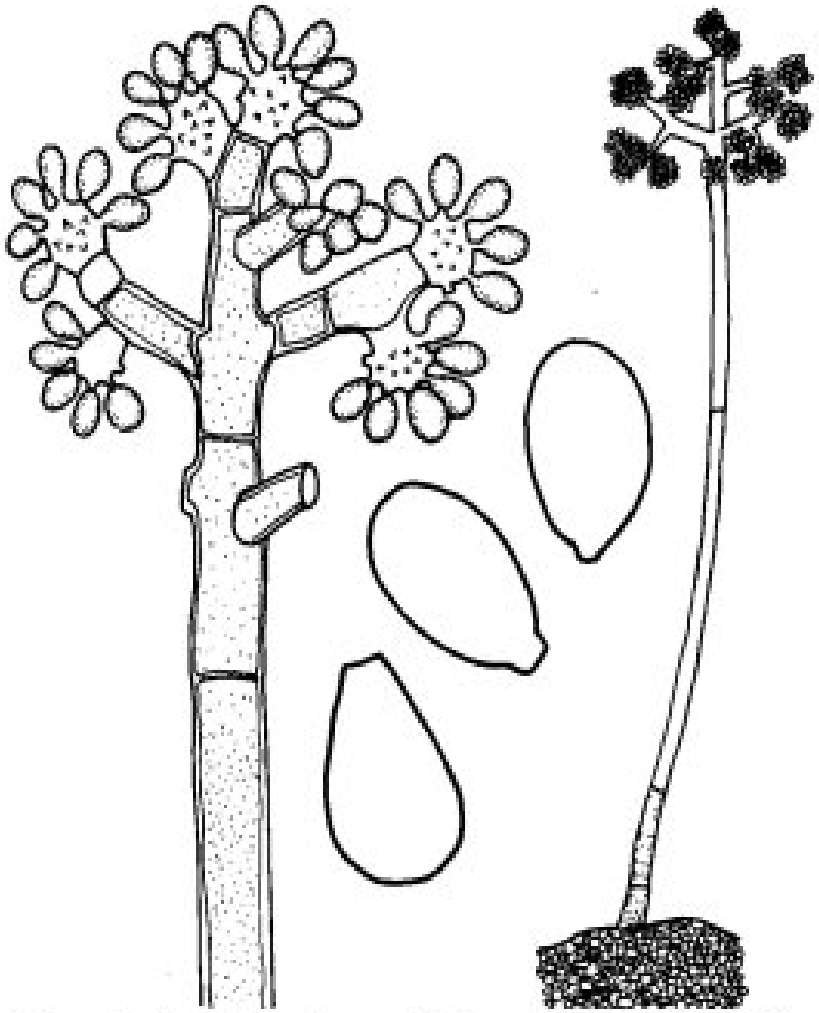


Figure 8. Condidiophores (left and right) and conidia (center) of Botrytis cinerea as you would see them under a high-power microscope.

Grey mold - *Botrytis cinerea*

Disease Cycle

- Botrytis blight is **a disease of greenhouse-grown chrysanthemum**.

  PSI: The fungus survives in plant debris as **sclerotia.**

  SSI: Spores are spread by overhead irrigation and by air.

Epidemiology

- The disease is favoured by high humidity and temperatures of 50°F to 70°F. **Warm days followed by cool nights encourages condensation of water on flowers** also favours disease development.

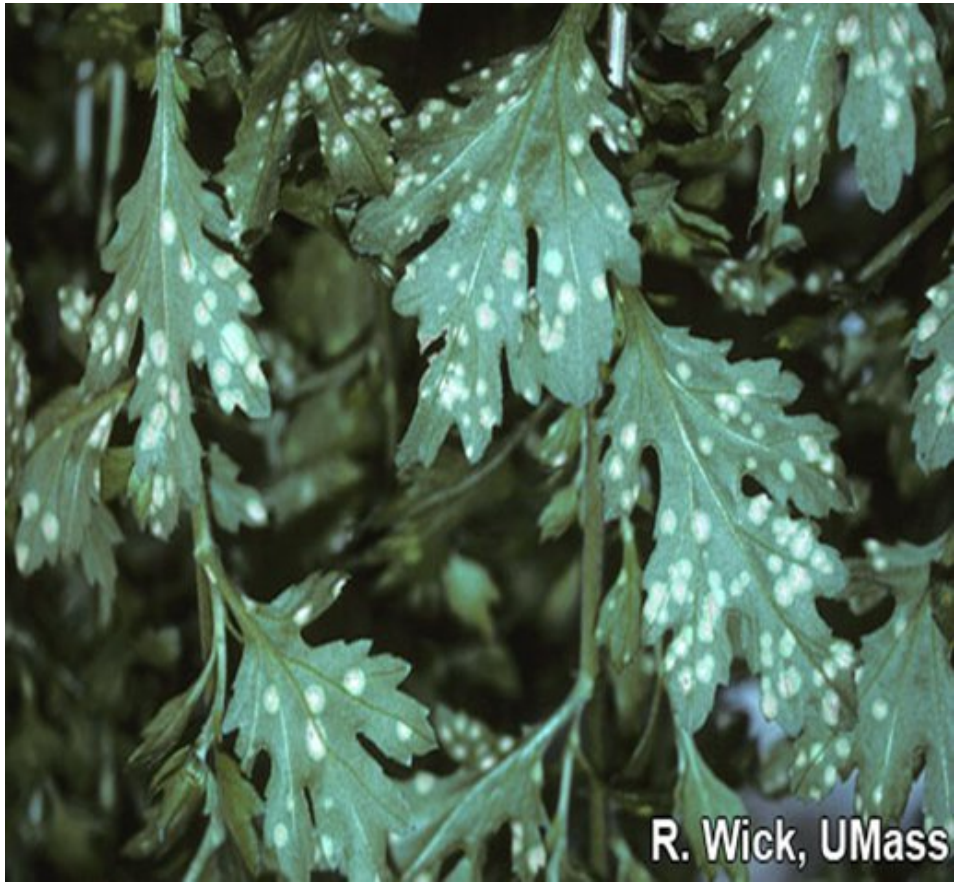
- **Continuous wetness for 5 to 8 hours** is necessary for conidial infection.

Grey mold - *Botrytis cinerea*

Management

- Practice good sanitation
- Use disease-free cuttings
- Regulate relative humidity
- Use adequate plant spacing to promote good air circulation
- Fungicides: Iprodione, thiophanate-methyl, thiophanate-methyl plus iprodione, mancozeb, chlorothalonil
- Biocontrol *Trichoderma harizanum* T-22

Rust - *Puccinia chrysanthemi, P. horiana*



Rust - *Puccinia chrysanthemi, P. horiana*

P. horiana **causes Chrysanthemum white rust (CWR) and as a recent introduction to the United States is subject to quarantine and an eradication program.**

Rust - *Puccinia chrysanthemi, P. horiana*

Rust pustules on lower surface

Pale, raised spots upper surface

Rust - *Puccinia chrysanthemi, P. horiana*

Rust - *Puccinia chrysanthemi, P. horiana*

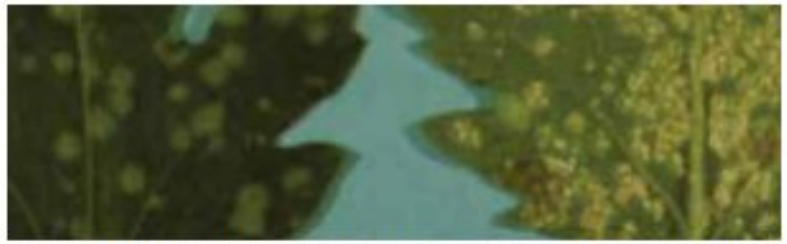


Figure 1. Upper and lower leaf surfaces with mature CWR infection.

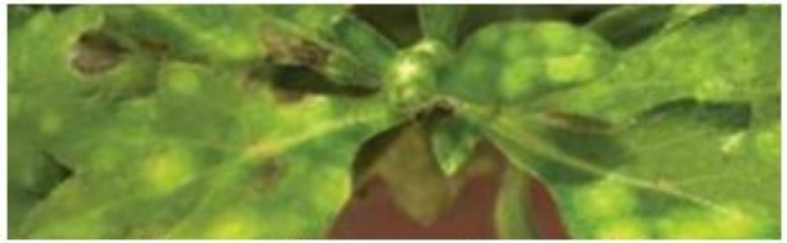


Figure 2. Yellow spots (first symptoms) on upper leaf structure.

Figure 3. Pinkish or buff-colored pustules on lower leaf surface.

Figure 4. Waxy white pustule on lower leaf surface.

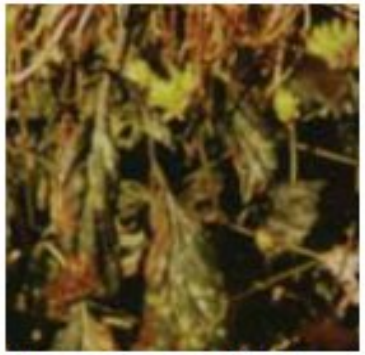


Figure 5. Severe CWR in a finished crop.

Etiology

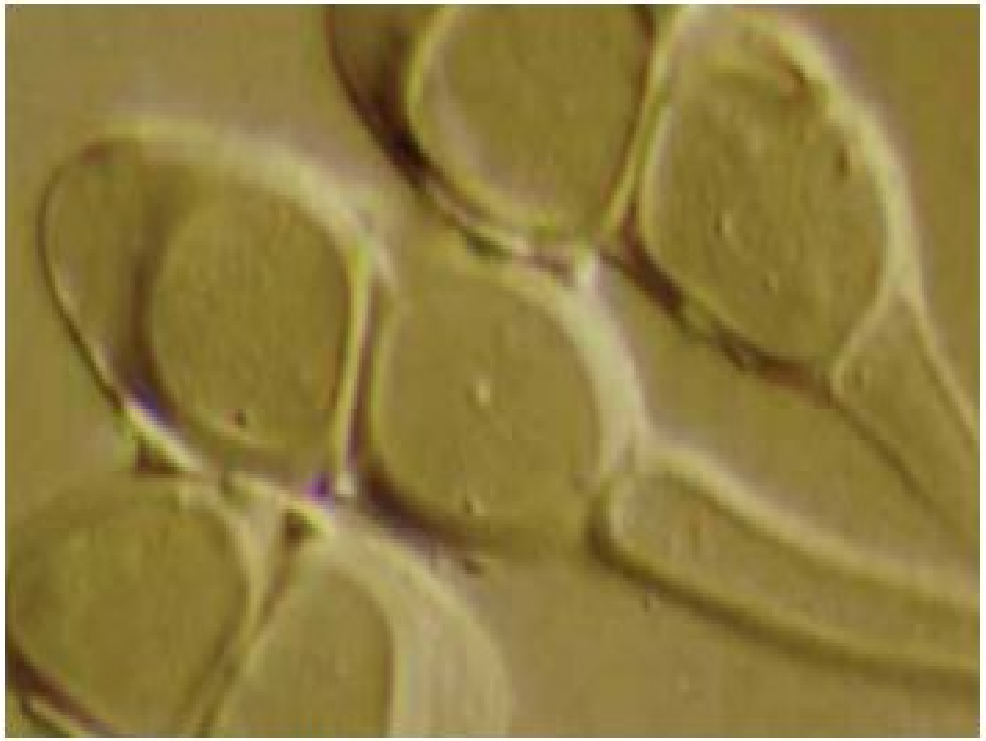


Figure 6. Teliospores of Puccinia horiana.

Rust - *Puccinia chrysanthemi, P. horiana*

- Two species of Puccinia causes rust on chrysanthemum

  P. chrysanthemi (Brown rust) ***P. horiana*** (White rust)

Brown rust **does not produce raised pustules** on the leaf lower surface; dark brown spores occur in powdery masses.

White rust fungus causes **brownish, pinkish, pale buff, or white spore** producing pustules on leaf lower surface.

Brown rust - ***P. chrysanthemi*** is most common in late summer

- **Brown rust** causes **minor damage in the field** and is **uncommon on greenhouse plants.**

- Symptoms as **dirty-brown pustules and yellowish-green spots on the lower surface of the leaves**

- Severe infestation may damage large areas of **leaves and lead to defoliation and reduced flower production.**

Rust - *Puccinia chrysanthemi, P. horiana*

white rust - *P. horiana* - in green houses

- Symptoms include **white, pinkish or brownish pustules produced on underside of leaves** with **white, yellow, to pale-green lesions on upper leaf surface.**
- Pustules **are most common on young leaves** may form on any green tissue, **flower bracts** or **the petals**.
- Chrysanthemum white rust results in leaf distortion, discolouration, defoliation, and plant death.
- **White rust is primarily a disease of greenhouse crops**; **when it occurs outside, direct sunlight and low humidity kill the spores.**

Rust - *Puccinia chrysanthemi, P. horiana*

Life cycle

- Chrysanthemum white rust has a **very short life cycle and only two spore stages.**
- **Reproduces only on living chrysanthemum foliage**.
- Teliospores develop on the underside of infected leaves and can persist up to 8 weeks on pustules in leaves.
- Germinate in the leaf pustules, producing basidiospores under cool, moist conditions (96-100% relative humidity for at least three hours).
- Basidiospores are fragile and short-lived (1 hour or less depending on the humidity) and **are responsible for explosive epidemics under congenial conditions**.

Rust - *Puccinia chrysanthemi, P. horiana*

Epidemiology

- Both brown rust and white rust of chrysanthemum are favored by the same conditions (cool temperatures and damp weather or high humidity).
- Infection can occur when temperatures are cool (between 40-73°F with an optimal temperature of 63°F) and relative humidity is high (96-100%).
- **A film of free water on the leaves is required for infection.**
- Symptom development usually occurs in 5 to 14 days.

Rust - *Puccinia chrysanthemi, P. horiana*

Management:

- Chrysanthemum varieties resistant to rust include 'Achievement', 'Copper Bowl', 'Escapade', 'Helen Castle', 'Mandalay', 'Matador', 'Miss Atlanta', 'Orange Bowl', and 'Powder Puff.
- **Infected plants to be destroyed to prevent disease establishment**

Fusarium wilt - *Fusarium oxysporum* f. sp. *chrysanthemi*

Fusarium wilt - *Fusarium oxysporum* f. sp. *chrysanthemi*

Fusarium wilt - *Fusarium oxysporum* f.sp. *chrysanthemi*

Symptoms

- Disease is characterized by chlorosis of one or more leaves
- Drooping of leaves and wilting of the affected plants
- Black necrotic spots appear on the stem
- Vascular discolouration

Fusarium wilt - *Fusarium oxysporum* f. sp. *chrysanthemi*

Survival and spread

- **P.I:** Soil borne fungus or through cuttings
- **S.I:** Conidia dispersed through irrigation water

Management

- Cuttings should be obtained from disease free plants
- Plant in pasteurized soil or soilless mix free of the pathogen.
- Maintain soil pH between 6.5 and 7.0 by using nitrate fertilisers rather than ammonium form of fertilizer.
- Apply a fungicide to protect healthy plants.
- Long crop rotation

Bacterial blight - *Erwinia chrysanthemi*

Bacterial Blight and Soft Rot [*Dickeya chrysanthemi* (syn. *Erwinia chrysanthemi*)]; [*Pectobacterium carotovorum* (syn. *Erwinia carotovora*)]

Bacterial blight - *Erwinia chrysanthemi*

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Bacterial blight - *Erwinia chrysanthemi*

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Bacterial blight - *Erwinia chrysanthemi*

Reddish brown rot

Bacterial blight - *Erwinia chrysanthemi*

Bacterial Blight and Soft Rot [*Dickeya chrysanthemi* (syn. *Erwinia chrysanthemi*)]; [*Pectobacterium carotovorum* (syn. *Erwinia carotovora*)]

Symptoms

- Cuttings **turn dark brown and collapse**
- **Surviving cuttings may be infected** but without symptoms
- In established plants **rotting of upper part of stem**
- Occasionally **marginal leaf scorching**
- Wilting and collapse of plants wilt during day and recover at night

Bacterial blight - *Erwinia chrysanthemi*

Bacterial Blight and Soft Rot [*Dickeya chrysanthemi* (syn. *Erwinia chrysanthemi*)]; [*Pectobacterium carotovorum* (syn. *Erwinia carotovora*)]

Survival and Spread

- Infected plants, soil, weed host.
- Rain drops

Favourable conditions

- High humid weather

Bacterial blight - *Erwinia chrysanthemi*

Bacterial Blight and Soft Rot [*Dickeya chrysanthemi* (syn. *Erwinia chrysanthemi*)]; [*Pectobacterium carotovorum* (syn. *Erwinia carotovora*)]

Management

- Destruction of affected plants, soil sterilization
- Select disease free cuttings
- Cutting dipped in Antibiotic solution for 4 hour such as streptomycin (Agrimycin)
- Spray of Streptocycline (0.01%)

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Fig. 1 Typical leaf symptoms induced by CChMVd variant CM5 on chrysanthemum cv. 'Bonnie Jean' (right), compared with a healthy control of the same cultivar (left).

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

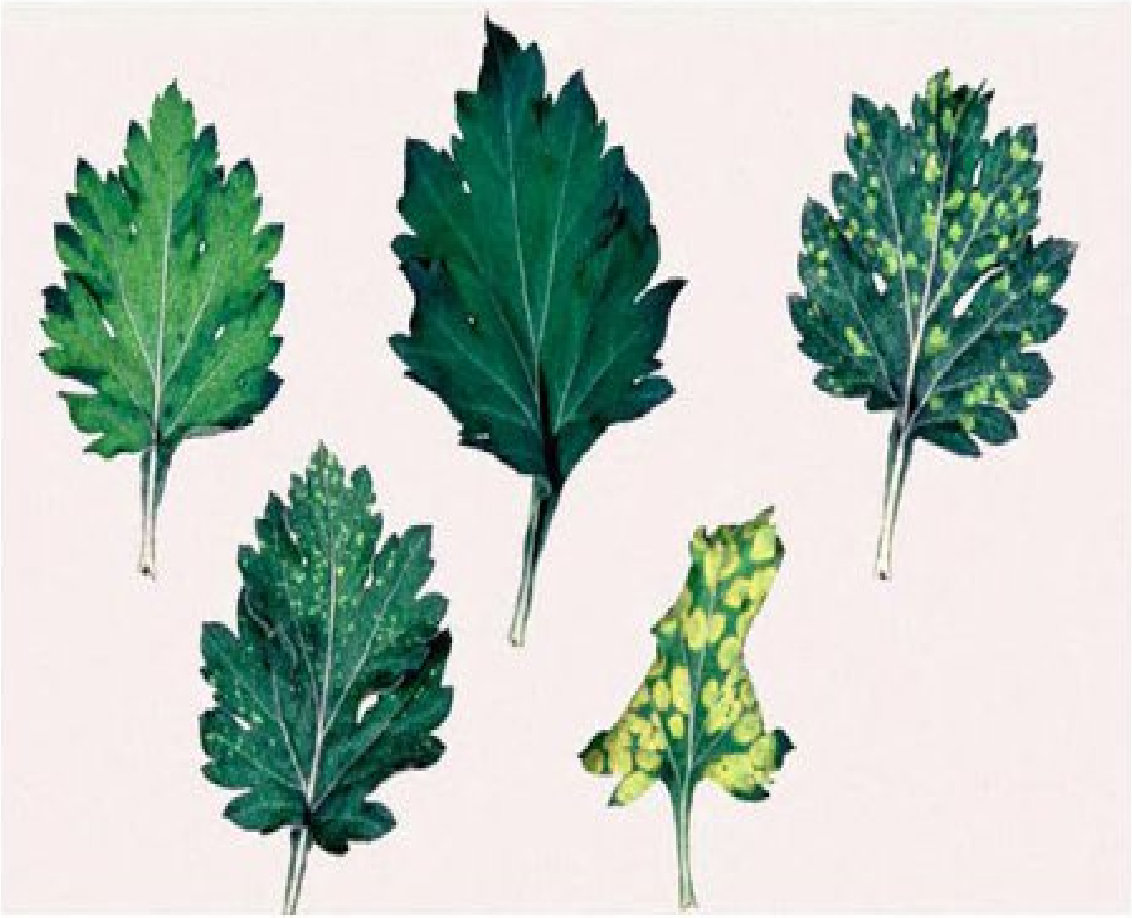

Plate 6 Foliar Symptoms on Chrysanthemum (Variety Bonnie Jean) caused by (clockwise from upper left) Chrysanthemum Chlorotic Mottle Viroid, Healthy Leaf, Potato Spindle Tuber Viroid, Chrysanthemum Stunt Viroid, and Potato Spindle Tuber Viroid (Mild Strain)

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Symptoms

- It causes **yellow-green mottling** followed by **complete chlorosis** of leaves with chronic symptoms
- Deforming of leaves, stems, and flowers.
- **Delay in flower** development
- Leaves may be marked with **ring spots, lines, pale areas or mottling**.
- Infected plants are **stunted form dense rosettes and have small flowers**

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Chrysanthemum chlorotic mottle viroid (***CChMVd***)

Family : *Avsunviroidae*

Genus : *Pelamoviroid*

Transmission: Sap, grafting

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Survival and spread

Primary:

- Viroid surviving in infected plants and spread through planting material and by vegetative propagation

Secondary:

- Viroid Spread through sap and knives or tools used during pruning or pinching plants and cutting the flowers

Chlorotic mottle - *Chrysanthemum Chlorotic Mottle Viroid* (*CChMVd*)

Management

- Remove and destroy infected plants
- Effective control can be done by using of viroid-free propagation material.
- Meristem-tip culture and subsequent heat treatment failed to eliminate viroid from infected cuttings
- Spread occurs readily by mechanical means, including plant-to-plant contact, handling during cultivation and contaminated cutting tools
- Good plant hygiene measures (e.g. use of disposable blades and gloves when taking cuttings, removal of dead material etc) can help to limit spread

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

3 Stunt on Chrysanthemum caused by Chrysanthemum Stunt Viroid. Var. Dark Pink Orchard

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Typical Mum Stunt symptom—shorter plants, smaller leaves, response variation. ***Not a Yoder Brand Variety.***

Typical Mum Stunt symptom—shorter plants, smaller flowers. ***Not a Yoder Brand Variety.***

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Symptoms

- Plants are small with paler leaves and flowers
- Plants are stunted to half their normal height at maturity.
- Axillary buds grow prematurely and **produce number of branches and stolons that** give rise to stunted plants
- Young leaves are **light green and very upright**.
- Some flowers appear **bleached and are inferior in quality**
- In diseased plants flowers open in 7 to 10 days earlier than healthy plants
- White specks and yellow blotches appear on leaves of some varieties
- Poor rooting in cuttings from diseased plants

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Family: *Pospiviroidae* **Genus**: *Pospiviroid*

Species: *Chrysanthemum stunt viroid* **Acronym**: *CSVd*

- Chrysanthemum Stunt viroid can survive at 212°F (100° C) for 10 minutes.

Transmission : Sap, grafting, and seed

Chrysanthemum stunt - *Chrysanthemum Stunt Viroid* (*CSVd*)

Survival and spread

- Chrysanthemum Stunt viroid is highly mechanically transmissible.
- **P.I:** Viroid survives from season to season in infected plants and plant debris, **remains infective in dried leaves as long as two years**
- **S.I: Spread through sap and knives** or tools used during pruning or pinching plants, taking cuttings and cutting the flowers

Management

- Use certified viroid free stocks
- Remove and destroy infected plants